

9.1.11.

Suppose that a coin is tossed three times and the side showing face up on each toss is noted. Suppose also that on each toss heads and tails are equally likely. Let HHT indicate the outcome heads on the first two tosses and tails on the third, THT the outcome tails on the first and third tosses and heads on the second, and so forth.

- a. List the eight elements in the sample space whose outcomes are all the possible head-tail sequences obtained in the three tosses.
- b. Write each of the following events as a set and find its probability:
 - (i) The event that exactly one toss results in a head.
 - (ii) The event that at least two tosses result in a head.
 - (iii) The event that no head is obtained.

a. List the eight elements in the sample space whose outcomes are all the possible head-tail sequences obtained in the three tosses.

Let S be the set of possible head-tail sequences, then:

$$S = \{HHH, HHT, HTT, TTT, THH, TTH, HTH, THT\}$$

Solution: a. $S = \{HHH, HHT, HTT, TTT, THH, TTH, HTH, THT\}$

b. Write each of the following events as a set and find its probability:

- (i) The event that exactly one toss results in a head.

Let E_0 be the set of event that exactly one toss results in a head, then:

$$E_0 = \{HTT, TTH, THT\}$$

Since $E_0 \subseteq S$, and S is a finite sample space in which all outcomes are equally likely, and E_0 is an event in S , then the probability of E_0 , denoted $P(E_0)$, is:

$$P(E_0) = \frac{\text{the number of outcomes in } E_0}{\text{the total number of outcomes in } S}$$

Let $N(E_0)$ = the number of outcomes in E_0 , and let $N(S)$ be the total number of outcomes in S . Then:

$$\begin{aligned}N(E_0) &= 3 \\N(S) &= 8 \\P(E_0) &= \frac{N(E_0)}{N(S)} = \frac{3}{8} = 0.375\end{aligned}$$

Hence, the probability of E_0 is 37.5%.

Solution: b. (i) The probability of exactly one toss results in a head is 37.5%.

(ii) The event that at least two tosses result in a head.

Let E_1 be the set of event that at least two tosses result in a head, then:

$$E_1 = \{HHH, HHT, THH, HTH\}$$

Since $E_1 \subseteq S$, and S is a finite sample space in which all outcomes are equally likely, and E_1 is an event in S , then the probability of E_1 , denoted $P(E_1)$, is:

$$P(E_1) = \frac{\text{the number of outcomes in } E_1}{\text{the total number of outcomes in } S}$$

Let $N(E_1)$ = the number of outcomes in E_1 , and let $N(S)$ be the total number of outcomes in S . Then:

$$\begin{aligned}N(E_1) &= 4 \\N(S) &= 8 \\P(E_1) &= \frac{N(E_1)}{N(S)} = \frac{4}{8} = \frac{1}{2} = 0.5\end{aligned}$$

Hence, the probability of E_1 is 50%.

Solution: b. (ii) The probability of at least two tosses result in a head is 50%.

(iii) The event that no head is obtained.

Let E_2 be the set of event that no head is obtained, then:

$$E_2 = \{TTT\}$$

Since $E_2 \subseteq S$, and S is a finite sample space in which all outcomes are equally likely, and E_2 is an event in S , then the probability of E_2 , denoted $P(E_2)$, is:

$$P(E_2) = \frac{\text{the number of outcomes in } E_2}{\text{the total number of outcomes in } S}$$

Let $N(E_2)$ = the number of outcomes in E_2 , and let $N(S)$ be the total number of outcomes in S . Then:

$$N(E_2) = 1$$

$$N(S) = 8$$

$$P(E_2) = \frac{N(E_2)}{N(S)} = \frac{1}{8} = 0.125$$

Hence, the probability of E_2 is 12.5%.

Solution: b. (iii) The probability of no heads is 12.5%.